

Appendix II

Code

Code for Servo movement with respect to EMG signals:

```
#include <Servo.h>
Servo servo1;
Servo servo2;
Servo servo3;
Servo servo4;

const int emgPin = A0; // EMG sensor connected to analog pin A0

const int servoPin1 = 9;
const int servoPin2 = 10;
const int servoPin3 = 11;
const int servoPin4 = 6;

void setup() {
  Serial.begin(9600);
  servo1.attach(servoPin1);
  servo2.attach(servoPin2);
  servo3.attach(servoPin3);
  servo4.attach(servoPin4);
}

void loop() {
  int emgValue = analogRead(emgPin);
  Serial.print("EMG Value: ");
  Serial.println(emgValue);

  if (emgValue <= 450) {
    servo1.write(0);
    servo2.write(180);
    servo3.write(0);
    servo4.write(0);
  }
}
```

```
else if (emgValue ==) {
    servo1.write(0);
    servo2.write(180);
    servo3.write(0);
    servo4.write(0);
} else {
    servo1.write(180);
    servo2.write(0);
    servo3.write(90);
    servo4.write(180);
}
delay(785);
}
```

Code for NTC thermocouple temperature sensor:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <max6675.h>
#include <SPI.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
#define SCK 18
#define CS 5
#define SD 19
MAX6675 thermocouple(SCK, CS, SD)
void setup() {
    Serial.begin(115200);
    if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
        Serial.println("OLED not found!");
        while (1);
    }
}
```

```

display.clearDisplay();
display.setTextSize(1);
display.setTextColor(WHITE);
display.setCursor(10, 10);
display.println("Initializing...");
display.display();
delay(2000);
Serial.println("MAX6675 Temperature Sensor Test");
}
void loop() {
  float totalTemp = 0;
  int readings = 3;
  for (int i = 0; i < readings; i++) {
    float tempC = thermocouple.readCelsius();
    if (tempC < -1) {
      Serial.println("Error: No Thermocouple Detected!");
      display.clearDisplay();
      display.setTextSize(2);
      display.setCursor(10, 20);
      display.println("No Sensor!");
      display.display();
      delay(2000);
      return;
    }
    Serial.print("Reading ");
    Serial.print(i + 1);
    Serial.print(": ");
    Serial.print(tempC);
    Serial.println(" °C");
    totalTemp += tempC;
    delay(1000); // 1-second delay between readings
  }

  float avgTemp = totalTemp / readings; Serial.print("Average Temperature: ");
  Serial.print(avgTemp);
  Serial.println(" °C");
  display.clearDisplay();
  display.setTextSize(2);
  display.setCursor(10, 10);
  display.println("Avg Temp:");
  display.setCursor(30, 35);

```

```
display.print(avgTemp, 1);
display.print(" C");

if (avgTemp > 45) {
    display.setTextSize(1);
    display.setCursor(10, 55);
    display.print(" 🔥 Danger: Burn Risk!");
    Serial.println(" 🔥 Warning: High Temperature! Burn Risk!");
}
else if (avgTemp < 10) {
    display.setTextSize(1);
    display.setCursor(10, 55);
    display.print(" ❄️ Warning: Too Cold!");
    Serial.println(" ❄️ Warning: Very Low Temperature!");
}
display.display();
delay(5000);
}
```